

ASHRAFUL FERDOUS RAFSUN

Aerospace Engineering Aspirant | STEM Researcher | Space & Aviation Projects | Computational & Data-Driven Problem Solving

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PROFESSIONAL SUMMARY

Motivated STEM student with a strong interest in Aerospace Engineering, Computational Science, and Space Technology, with hands-on experience developing engineering and technology projects using programming and data-driven approaches. Experienced in building projects related to aircraft performance, rocket trajectory simulation, satellite tracking, and space missions, while independently exploring aerospace research and aerodynamic analysis. Strong foundation in Mathematics, Physics, Chemistry, and Biology, complemented by growing skills in programming, research, and technical problem-solving. Seeking opportunities to contribute to STEM research, engineering projects, and international academic programs while pursuing a future career in Aerospace Engineering.

EDUCATION

Uttarkhan Collegiate School | Dhaka 2026
Secondary School Certificate (SSC) CGPA : 3.50

• Uttarkhan Collegiate School Dhaka

Uttara High School And College | Dhaka, Bangladesh 2026
Higher Secondary Certificate (HSC)

PROJECTS

AeroPerf – Flight Performance Calculator | GitHub Jul 2026 - Present

Front-End Developer

- Built an aviation-focused calculator for runway performance, MTOW, density altitude, wind, and safety calculations. Designed a modern responsive interface using **HTML, CSS, and JavaScript**.

Technologies / Tools Used : HTML5, CSS3, JavaScript

AstroLaunch – Live Space Launch Tracker | GitHub Jul 2026 - Present

Front-End Developer

- Developed a **responsive web app** to display live space launch data using REST APIs.
- Integrated real-time mission details and countdown functionality with **dynamic JavaScript**.

Technologies / Tools Used : HTML5, CSS3, JavaScript

ISS Live Satellite Tracker | GitHub Jul 2026 - Present

Front-End Developer

- Developed a **web application** to track the International Space Station (ISS) in real time.
- Integrated live satellite location data using REST APIs and **dynamic JavaScript**.

Technologies / Tools Used : HTML5, CSS3, JavaScript

Rocket Trajectory Simulator | GitHub Jul 2026 - Present

Front-End Developer

- Developed an **interactive rocket trajectory simulation** using **JavaScript**.
- Implemented mathematical calculations to visualize projectile motion and flight paths.

Technologies / Tools Used : HTML5, CSS3, JavaScript

NASA Mission Control Dashboard | GitHub Aug 2026 - Present

Front-End Developer

SKILLS

- Developed a **mission control dashboard** to display NASA mission and space-related data.
- Integrated REST APIs to fetch and present dynamic mission information.

Technologies / Tools Used : HTML5, CSS3, JavaScript

Languages :	Bengali, English
Programming Languages :	HTML5, CSS3, JavaScript
Soft Skills :	Problem Solving, Critical Thinking, Research, Communication, Teamwork
Tools & Platforms :	Git, GitHub, GitHub Pages

EXTRACURRICULAR ACTIVITIES

School & Local Soccer Player <i>Represented the school football team</i> Actively played for a local club, and won a local tournament championship.	Mar 2022 - Present
RAJUK National SciSpark 4.0 <i>Presented a STEM project</i> Received a participation certificate.	June 2026
Photography <i>Building a portfolio</i> Developing visual storytelling and composition skills.	Jan 2023 - Present
Aquarium Ecosystem Management <i>Independent Hobby</i> - Maintain a freshwater aquarium by monitoring pH, ammonia, nitrite, nitrate, chlorine, and temperature. - Apply water chemistry and biological filtration principles to maintain a stable aquatic ecosystem. - Troubleshoot water quality issues through testing and preventive maintenance.	Feb 2024 - Present

AWARDS & CERTIFICATIONS

Best Student Performance Award 2026 — ALCCO–Cambrian Education Group Recognized for outstanding performance in the SSC/Equivalent Examination 2026 with a GPA of 4.39	Aug-2026
EMU Project–2028 Scholarship — ALCCO–Cambrian Education Group Received the EMU Project–2028 Scholarship as part of the academic performance recognition.	Aug-2026
CS50: Introduction to Computer Science — Harvard University Foundations of computer science, programming, algorithms, and computational problem-solving.	Aug-2026
Aircraft Designing — Alison Fundamentals of aircraft design, aerodynamics, and aircraft performance.	June-2026
Artificial Intelligence — HP LIFE Fundamentals of artificial intelligence and its practical applications.	May-2026